Measuring the Dynamics of Beliefs on Social Media

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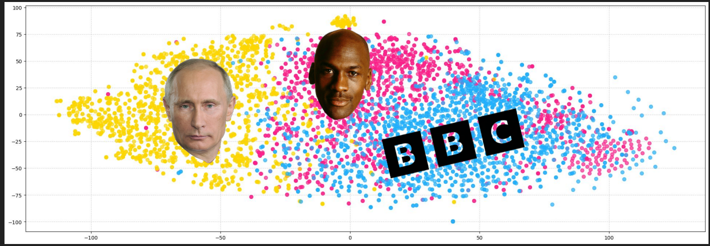
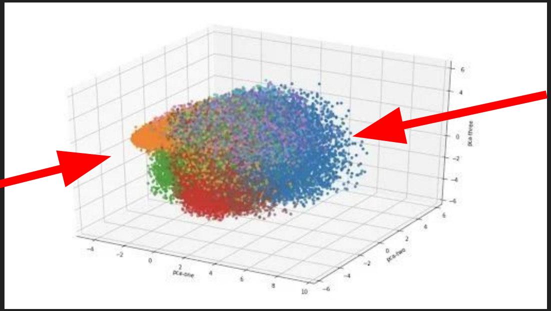
Research Question

- Research Question A:
  - How do tweets from various clusters differ in language and content, and can we visualize meaningful clusters?



- Research Question B:
  - Can map out how public opinion changes over time regarding controversial topic?

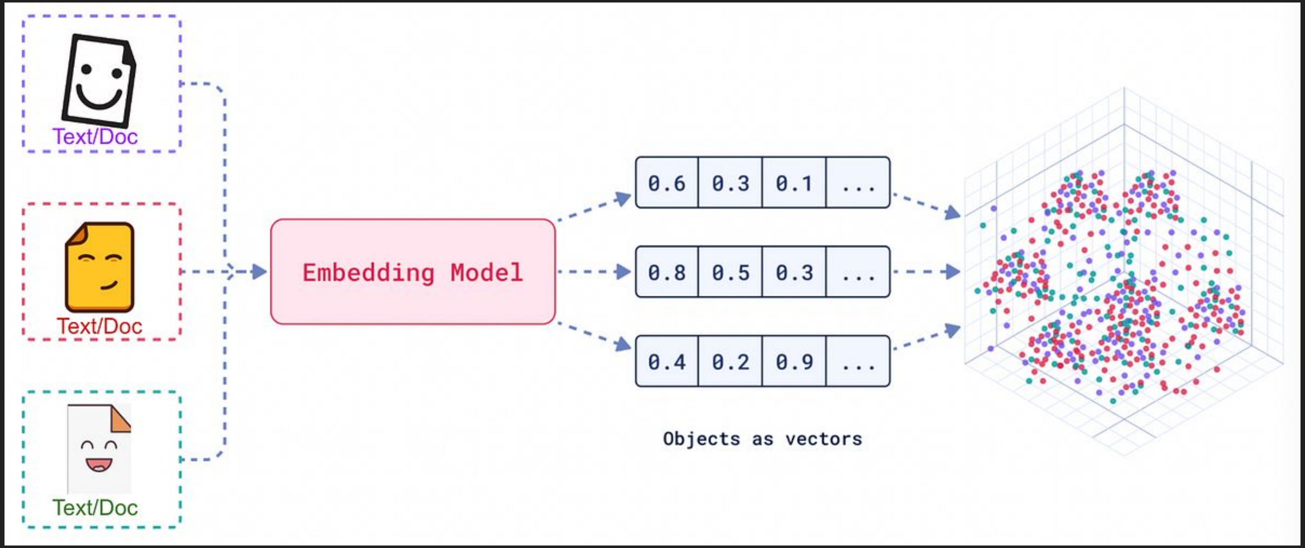


Method

1. Use a scraper to collect tweets regarding certain topics
2. Clean and format data
3. Use **text embedding** to represent each tweet as a numeric vector
4. Use dimension reduction (**t-SNE**) to reduce total features to 2
5. Cluster embedded tweets in 2D to visualize how themes and topics differ
6. Evaluate accuracy of clustering

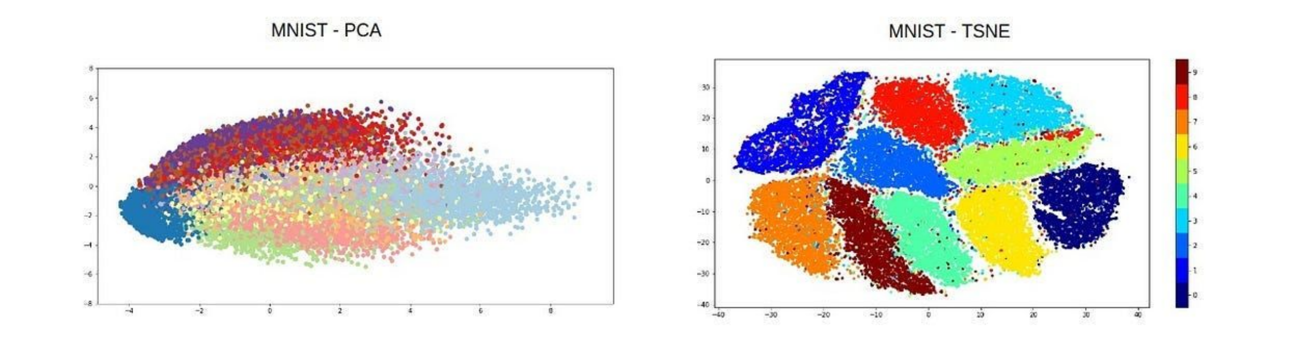
Background: Text Embedding

- Text Embedding is just a vector of n dimension (usually 1024 dimensions) that represent the idea/belief space of a given string of text
- We used the python package, sentence_transformer, to get this capability



What is t-SNE?

- A way to reduce dimensions for visualization purposes
  - Typically reduces many dimensions down to just 2 or 3
- As opposed to PCA, t-SNE can capture non-linear relationships within the original data
- Main Strength: Points that are close together in the high-dimensional space will tend to be close in the t-SNE plot

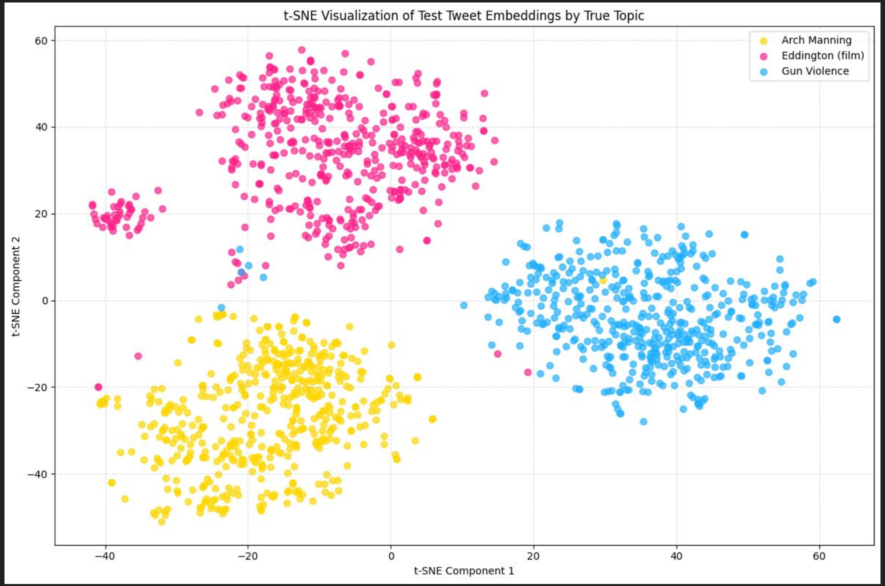


Dataset

- The dataset we used consists of tweets scraper from Twitter regarding certain topics spanning ~10 days in length
- We selected 3 very different topics, in order to hopefully visualize clear difference among clusters
    - Arch Manning (athlete)
    - gun violence (politics)
    - Eddington (film)

Visualizations

- After applying text embedding and dimension reduction (t-SNE), tweets (without labels) created clusters that appeared to be visibly different.



How do opinions change over time?

- Now that we know it is possible that different topics appear significantly different in 2D space, is it possible for us to map model the semantics of a topic changes over time?
  - Statics -> Dynamics
- For this, we used 1 topic: The recent US government shutdown

Polarization Index Theory

$$P = \alpha\mu_{gf} + \beta\mu_{gs} + \gamma\mu_{gb}$$

P = Polarization Index
μ_gf = Group Formation Dynamics Coefficient
μ_gs=Group Separation Quality Coefficient
μ_gb=Group Size Balance Coefficient
Alpha, Beta, Gamma = Normalization Coefficient

Theory: Group Formation Dynamics Coefficient

- We measure how many natural groups exist over time. Add more groups create polarization
- We use a consensus confidence algorithm to prevent the k number from fluctuating very rapidly from noisy data to ensure temporal stability
- Alpha and Beta are normalization parameters K is the normalized number of k cluster amount and C is the consensus confidence value

$$\mu_{gf}(K, C) = \alpha_{gf} K + \beta_{gf} C$$

Theory: Group Separation Quality Coefficient

- We measure how well separated and distinct each cluster are. The more separation and easy distinguishment, the higher the polarization.
- We use the text embedding to generate the silhouette score (X), inter/intra cluster separation ratio (Y), the v vector is the text embeddings
- Alpha and Beta are the normalization parameters

$$\mu_{gs}(\vec{v}) = \alpha_{gs} X(\vec{v}) + \beta_{gs} Y(\vec{v})$$

Theory: Group Size Balance Coefficient

- We measure how balanced of each group cluster are relative to each other.
- Ex: If there are many groups, but one very large group and ton of smaller groups, there is less overall polarization despite having many groups. Conversely, if there are many groups, but they are evenly sized, there is a lot of polarization due to fragmentation in beliefs.
- S is the Normalized Shannon Entropy, which is maximized when the number of data points in each cluster is the same, D is the dominance penalty, and the k vector is the assigned cluster labels.

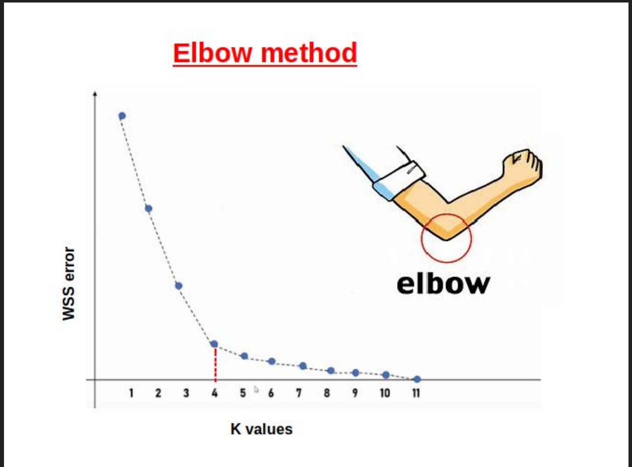
$$\mu_{gb}(\vec{k}) = \alpha_{gb} S(\vec{k}) + \beta_{gb} D(\vec{k})$$

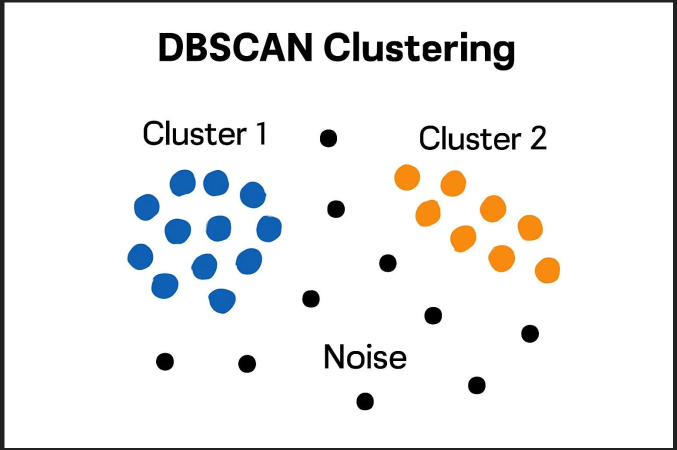
Theory: Rolling Window Frame Analysis

- Past discussion lose relevance over time
- Instead of analyzing the entire text embedding cluster globally, we create a bunch of time slices with text embedding representing the discussion context occurring at that time

Theory: How to determine n_clusters?

- We used a technique called consensus clustering that picked the k value that showed up the most amount of time from different techniques
- These were the method that we checked against: Elbow Method (Second Derivative test), Silhouette Score, Calinski-Harabasz Index, BIC for Gaussian Mixture Model, and HDBSCAN





Theory: Temporal Stability

- The index algorithm without any temporal stability is susceptible to noise when determining n_cluster value
- As an example, there is three factions inside a debate. And over time one faction may split off into two. It is very unlikely that the one factions will suddenly split into 12 factions.

SOLUTION: CONSENSUS CONFIDENCE SCORE

Methodology Validation

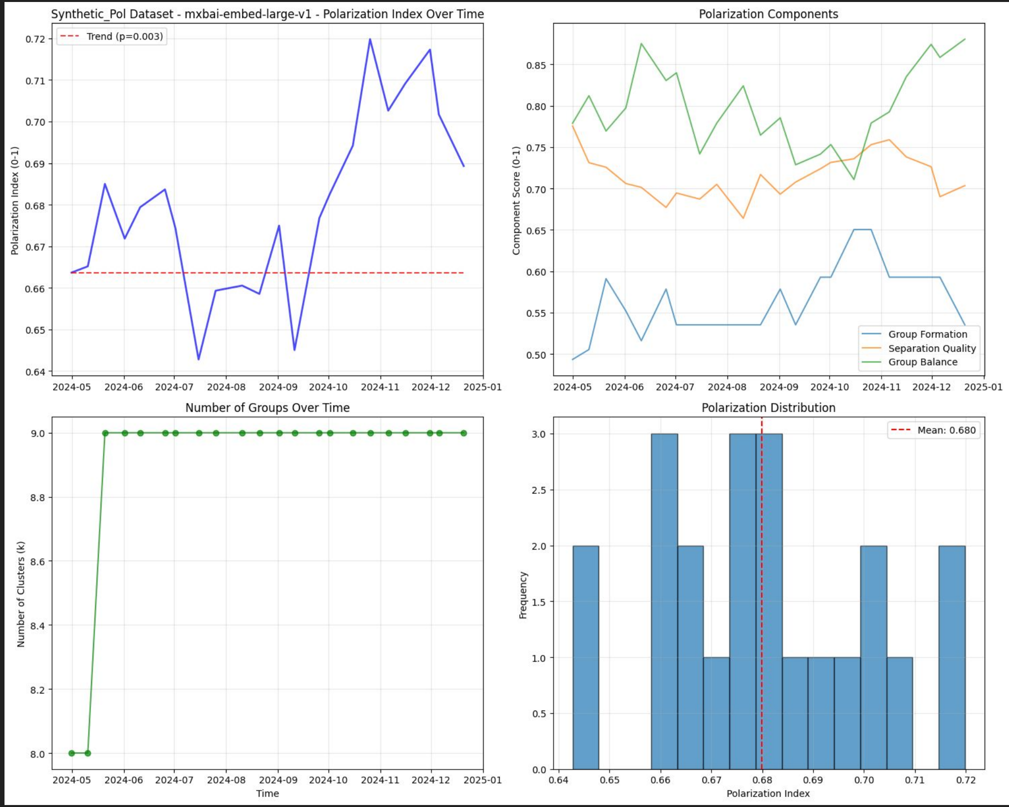
- Let Alpha=0.4, Beta=0.3, and Gamma=0.3
- **Echo Chamber**: One Dominant Group and One Small Group, **Binary Polarization**: Two equally Sized Group, **Multi-Polarization**: Four Different Equally Sized Group, and **Fragmentation**: 7 Equally Sized Group that are spread apart

Condition	Polarization Index Value	Formation	Separation	Balance
Echo Chamber	0.495	0.33	0.76	0.45
Binary Polarization	0.676	0.33	0.81	1.00
Multi Polarization	0.736	0.43	0.89	1.00
Fragmentation	0.780	0.57	0.84	1.00

Application: Measuring polarization in political debate dataset

- We use a synthetic dataset of a political debate dialog occurring over multiple people generated by Anthropic CLAUDE and a real dataset from Twitter that was scrapped about the recent US government shutdown that lasted for 45 days
- This synthetic dataset has 51 topics ranging from: healthcare, gambling, nuclear and etc
- We used the *mixedbread-ai/mxbai-embed-large-v1* model which achieved archives state of the art performance of Google’s Bert-large sized models in 2024
- We expect that the algorithm should map out the polarization index over time for a discussion

Results and Discussion: Synthetic Data



Results and Discussion: Twitter Data

Limitations

- Normalization Issue: The weights need to be determined. Different situations require different weights which doesn’t generalize across all situations.
- Mathematical Bounding: The index is not guaranteed to be between [0,1]
- Rolling Window Analysis: We considered the groups for a given time frame, there might be a way to use the global frame to make our algorithm more robust.